Research Article

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Robust baseball pitch reconstruction using artificial neural networks with noisy data

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Abstract: The use of high-fidelity camera-vision and Doppler tracking systems in Major League Baseball (MLB) has created an influx of advanced analytics that have transformed game and personnel strategy. However, due to the cost and complexity of these systems, advanced statistics are severely limited in most amateur games. In this work, we develop a robust pitch reconstruction methodology using artificial neural networks (ANN) and a three-point reconstruction technique, requiring the knowledge of only three baseball spatiotemporal locations. The ANN models were trained to predict the initial baseball speed and spin rate, which are then used as initial conditions to integrate the full pitch trajectory. We use numerically simulated baseball pitches to train two ANN models, each with clean and noisy training inputs, respectively. We then performed a robustness analysis to test ANN performance with increasingly noisy data to simulate low-fidelity camera tracking systems. Probability distributions of predicted model output are calculated using the Monte Carlo method to quantify model uncertainty. We show that the ANN models accurately predict the true baseball trajectory from both high-fidelity and noise-injected testing data. The results demonstrate the effectiveness of ANN models for quick and robust pitch trajectory reconstruction using minimal input data, even in the presence of noise. The present methodology provides a first step towards enabling pitch tracking and advanced analytics in a wider variety of baseball games.

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1 Introduction

Statistics have always been an integral part of sports, and baseball has always been on the cutting edge of advanced analytics. The game has changed dramatically with the onset of the digital revolution, culminating in entirely new philosophies on how to manage teams, select personnel, and train players by using traditional machine learning approaches, including classification and regression (Koseler and Stephan 2017), and Bayesian statistics (Deshpande and Wyner 2017; Santos-Fernandez et al. 2019). Today, it is impossible for any team to compete that is not taking advantage of the modern statistical approach to all facets of the game (Hintz 2022). Advanced analytics are only useful when there is quality data available, which is not often feasible at the amateur level. Thus, the development of robust tooling is necessary to provide the quantitative metrics that enable better decision making for personnel managers.

Nearly 20 years ago, Major League Baseball (MLB) introduced PITCHf/x, a computer vision system to track baseball trajectory in 3D (Healey 2017), consisting of an array of high-quality cameras. The enormous influx of high-fidelity in-game data allowed for teams to gain new insights and edges over the competition by utilizing advanced statistical methods (Hintz 2022). Detailed pitch tracking has transformed the dynamic between pitcher and hitter, as pitchers are working to improve their mechanics and quality of pitches while hitters study and train on representative pitches reconstructed from in-game pitch tracking. However, the widespread use of high-fidelity tracking systems is prohibitively expensive and has only trickled down from professional venues to high-level collegiate baseball, necessitating the need for low-cost alternatives. For example, this may involve an application that utilizes a standard smartphone camera (or array of smartphone cameras) and leverages robust reconstruction algorithms to quantify pitch trajectories. Such a low-cost system could be set up quickly and operated by any coach, player, or fan at any remote field. However, due to

this lack of technology and quantitative analysis, intensive scouting is still required at the amateur level, which is costly and error-prone (Weller 2020). The benefits of having access to high-fidelity data are clear from both a management and player perspective, allowing for the smart-training, development, and analysis of potential prospects.

Pitch tracking provides detailed data on pitch speed, movement, and spin, allowing coaches and players to identify strengths and weaknesses, refine mechanics, and optimize performance. Pitch tracking requires two major steps: 1) quantifying the baseball location from cameras and sensors, and 2) reconstructing pitch trajectory (initial position, velocity, and spin) using computational algorithms. For the former, a review of computer vision techniques in sports is given by Rahimian and Toka (2022) with emphasis on accuracy and cost-efficiency. Specifically, much effort has been undertaken to quantify the spatiotemporal location from high-speed cameras in order to reconstruct pitches in baseball (Shum and Komura 2004, 2005; Theobalt et al. 2004).

Recently, MLB has switched to a computer-vision optical system (provided by Hawk-Eye) for pitch tracking (An 2024), providing extremely precise spatiotemporal pitch information. Once the three-dimensional spatiotemporal baseball locations are determined, pitch reconstruction (step 2) is performed to determine velocity, spin rate, and pitch type. The most accurate and widely used methods for pitch reconstruction involve an iterative non-linear least squares fit between the measured pitch location and simulated pitch trajectory (Nathan 2007). The simulated pitches require a model for the surface and body forces acting on the ball during flight. The result is that each pitch in MLB is first tracked then reconstructed with high precision, allowing teams to improve and scout pitchers across the league.

Limited studies have considered cost-effective baseball tracking and pitch reconstruction methodologies. Recent work by Kim et al. (2019) showed that detailed pitch reconstruction can be achieved using unsynchronized cameras, reducing the sophistication required for future camera-tracking systems. Additionally, preliminary work by Hsieh (2024) showed that it is possible to extract the three-dimensional pitch location from a single two-dimensional viewpoint. For the second phase of pitch reconstruction, Aguirre-López et al. (Aguirre-López et al. 2018; Escalera Santos et al. 2019) showed that it is possible to reconstruct the entire pitch trajectory from just three spatiotemporal frames. The pitch trajectory can be reconstructed by minimizing the least-squares error between a simulated pitch and the three spatiotemporal frames, rather than using all frames over the entire pitch. This method was robust for high-fidelity systems with error less than 2 mm in baseball location, but has not been extended to lower-fidelity tracking systems prone to noise.

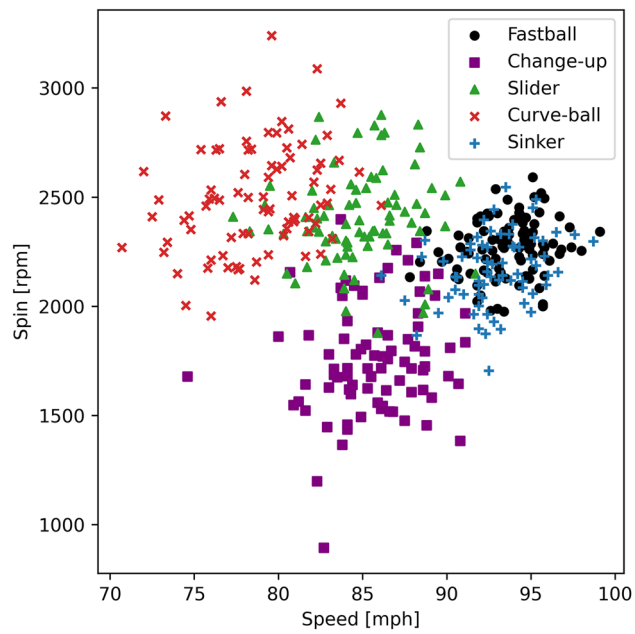


Figure 1: Average speed and spin rate classified by pitch for top 100 MLB pitchers in 2023 (Willman 2023).

Figure 1 shows the classification of pitch by average speed and average spin rate for the top 100 MLB pitchers in 2023 obtained from MLB's data analytics website (Willman 2023). The speed and spin rate (along with spin axis) are the parameters of most importance for determining pitch trajectory, and are used most often to classify pitch type. With variation, pitchers are able to switch up the baseball trajectory to keep batters guessing at the plate. Classification of pitch requires knowledge of the individual pitcher as the overlap between fastballs, sliders, sinkers, and change ups is hard to identify and would be taken care of by a separate machine learning model (Umemura et al. 2021). A suitable reconstruction algorithm needs to be applicable across this range of spin rate and velocity.

In the present work, we consider the novel use of artificial neural networks (ANNs) for trajectory reconstruction on synthetic pitches with variable speed and spin rate. The computer vision and tracking problem is not considered in the scope of this paper. The use of ANNs in baseball has been scarce relative to the proliferation of ANNs in other fields. ANNs in baseball have been used to predict winning games (Huang and Li 2021; Jeong et al. 2017), attendance of stadiums (Park and Park 2017; Park et al. 2018), pitch type and location (Lee 2022), and hall of fame status (Smith and Downey 2009; Young et al. 2008). ANNs (Zou et al. 2009) are considered to be universal function approximaters, mak-

ing them well suited to quickly and accurately capture the non-linear baseball trajectory from limited data inputs. Further, recent work with deep neural network models has shown that injection of random noise into the training dataset can improve the robustness of models (Fawzi et al. 2016; He et al. 2019; Sietsma and Dow 1991). In the present study, we train ANNs using the three-point approach, in which only three spatiotemporal frames of the baseball location are needed, representing the minimum viable information necessary to determine the entire pitch trajectory. We design ANN models that are suitable for low-fidelity camera tracking systems (providing noisy baseball location) by exploiting the robustness properties of ANN with injected noise. Ultimately, the development and proliferation of robust pitch tracking systems will be critical to expanding the accessibility of quality analytics.

Our novelties and contributions of this research are as follows:

- The proposed reconstruction framework has the potential to democratize access to quantitative data at all levels of baseball. Limited studies have considered pitch reconstruction in settings where high-end technology is inaccessible, which will enable the development of robust low-cost systems (e.g. smartphone applications) for collecting quantitative pitch metrics at all levels. The development of such systems is vital to bringing advanced analytics to amateur and international baseball to improve player performance, scouting, and personnel management.
- The use of ANN models for pitch reconstruction is the main novelty of this work. To our knowledge, the use of ANN has not been considered previously for trajectory reconstruction. ANN models require substantially less domain expertise to evaluate pitch trajectory in comparison to traditional iterative approaches, empowering continued development.
- By utilizing the robustness characteristics of ANNs trained on noisy data, our algorithm demonstrates greater robustness properties over existing methods. This proposed ANN-based model enables the development of cost-effective solutions for pitch tracking using noisy low-cost cameras and sensors. The robustness methodology considered will serve as a baseline for evaluating the performance of future algorithms and systems.

The rest of this study is organized as follows. Section 2 details the synthetic pitch generation and ANN model training. Section 3 presents the results, including a robustness analysis performed to determine the suitability of ANN prediction with increasing input noise. Section 4 draws conclusions, showing the ANN to be capable of reconstructing pitch trajectory with high accuracy and robust error properties.

2 Numerical method

2.1 Simulated baseball pitches

Once the baseball leaves the pitcher's hand, the complicated physics governing the aerodynamics of baseball flight determines the trajectory of the pitch. The pitcher chooses a pitch type (with a corresponding intended trajectory) and imparts the initial velocity and spin to the baseball. Figure 2 shows a schematic free-body diagram of the forces acting on the baseball during flight. Deflection downwards of the streamlines around the ball is due to rotation, which gives the characteristic lift force seen in balls with backspin (due to the magnus force). By invoking Newton's second law, the summation of forces acting on the baseball give the corresponding baseball acceleration during flight:

$$\mathbf{a} = \frac{\mathbf{F}}{m} = \mathbf{f}_D + \mathbf{f}_M + \mathbf{f}_G \tag{1}$$

where $\mathbf{f}_D$, $\mathbf{f}_M$, and $\mathbf{f}_G$ are the drag, magnus, and gravity acceleration due to their respective forces (note: bolded symbols denote vector quantities). Additional details of the governing equation, submodel selection, and discussion of model validity, are provided in Appendix A.

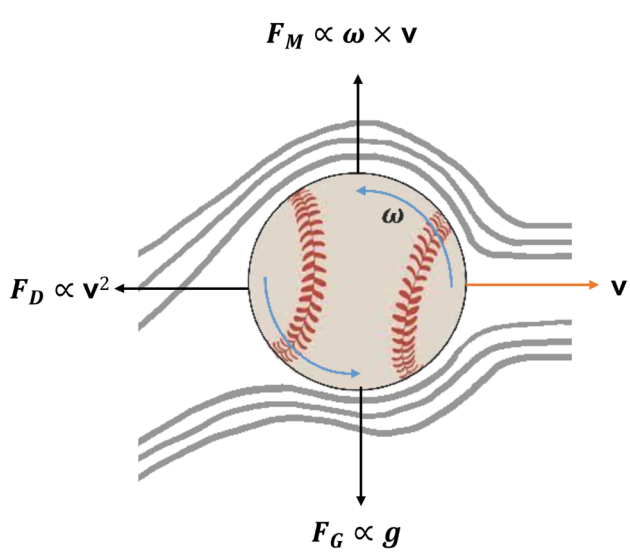


Figure 2: Free-body diagram of baseball forces during flight. Drag ($\mathbf{f}_D$), magnus ($\mathbf{f}_M$), and gravitational ($\mathbf{f}_G$) forces labeled along with baseball velocity, $\mathbf{v}$. Figure adapted from Escalera Santos et al. (2019).

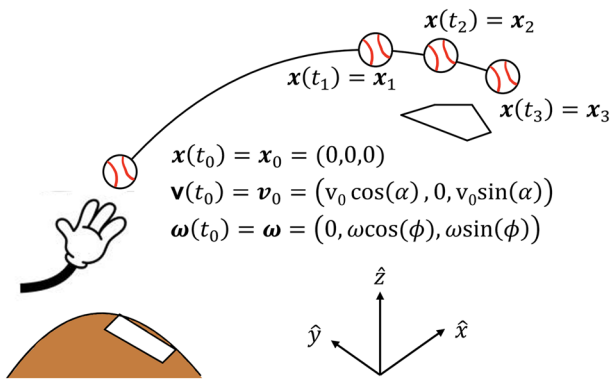


Figure 3: Schematic of pitch trajectory input to neural network using three-point reconstruction method. Initial pitch velocity, v_0, and spin rate, ω, are determined solely based on three spatiotemporal points: x_1, x_2, and x_3.

We simulated synthetic pitches in Python using a batch script. Figure 3 visualizes the simulated pitch schematic with the locations along the trajectory of the three spatiotemporal position frames $(\boldsymbol{x}_1, \boldsymbol{x}_2, \boldsymbol{x}_3)$ used for pitch input to the ANN for pitch reconstruction. The x-axis is oriented horizontally from the pitching rubber to home plate, the y-axis is oriented horizontally cross-wise towards the left-handed batters box, and the z-axis is oriented vertically up. Table 1 shows the pitch trajectory parameters for the batch run of synthetic baseball pitches, where $|\omega|$, v_o, ϕ, α, t_f, and Δt are the spin rate magnitude, initial velocity magnitude, spin axis angle, launch angle, simulation duration, and simulation time step, respectively. Based on data encapsulated in Figure 1 (Willman 2023), we varied the spin rate magnitude, $|\omega|$, between 800 and 3,300 rpm and the initial velocity magnitude, v_o, between 70 and 100 mph. We trained the ANN on samples outside the bounds of typical pitches to provide a robust sampling distribution (helpful in predictive performance for real world trajectories). We set the initial velocity as $\mathbf{v} = (\mathrm{v}_o \cos\alpha, 0, \mathrm{v}_o \sin\alpha)$, where α is 1°. The batch runs assumed a constant spin rate axis oriented along direction $\phi = 45°$ equidistant between the y- and z-axis with no component in the x-direction. We assumed the initial position to be $\boldsymbol{x}_o = (0, 0, 0)$ for all pitches. Additional discussion on the selection and validity of these assumptions is provided in Appendix A. We found a constant time step of $\Delta t = 2.083$ ms to be sufficiently accurate through a refinement study.

Table 1: Summary of pitch trajectory parameters used for batch synthetic pitch simulations: $|\omega|$, v_o, ϕ, α, t_f, and Δt are the spin rate magnitude, initial velocity magnitude, spin axis angle, launch angle, simulation duration, and simulation time step, respectively.

$\lvert\omega\rvert$ [rpm]	v_o [mph]	ϕ [deg]	α [deg]	t_f [s]	Δt [ms]
800–3,300	70–100	45	1	0.5	2.083

We produced synthetic data for training of the ANN by integrating the governing equations for baseball flight trajectory. We simulated 30,000 pitch trajectories with a randomly sampled spin rate and initial velocity from a uniform distribution. Using a large number of synthetic trajectories increases the robustness of model performance by supplying many samples along extreme trajectories (Fawzi et al. 2017). We compiled three position frames in the pitch trajectory from each trajectory at 60 frames per second apart, with the last frame corresponding to $t_3 = 0.5$s for all runs. We chose 60 frames per second as the interval rate as it is standard for commercial mobile phones. Given the pitch locations at $\mathbf{x}(t_1)$, $\mathbf{x}(t_2)$, and $\mathbf{x}(t_3)$, the three-point algorithm determines all initial conditions needed for pitch reconstruction. We compiled the trajectory locations, and initial spin rate and velocity into a dataset and stored as a.csv file to be used for training the artificial neural network to perform pitch reconstruction.

2.2 Artificial neural network architecture

We used the synthetically produced dataset (outlined in the previous subsection) to train an ANN model using the open-source TensorFlow (Abadi et al. 2016) library. The training methodology for the pitch tracking is summarized in Figure 4. We split the dataset into a training and testing set, with 80 % and 20 % of the data, respectively. One ANN was trained with no artificial noise in the training set, designated "**Quiet**", and one ANN was trained with a

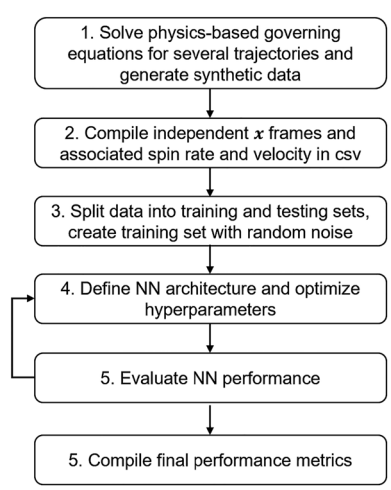


Figure 4: Schematic flow diagram of the training methodology for neural network pitch reconstruction. Neural network architecture is updated and optimized until performance reaches acceptable threshold.

uniform distribution of maximum error 25 mm (∼1 inch), designated "**Noisy**". We injected noise at each sample point independently from a uniform distribution with a known maximum error. The injection of random noise simulates the uncertainty associated with identifying spatiotemporal location of the baseball in a low-fidelity camera tracking system. After training, we injected the testing dataset with noise in 2 mm increments up to 50 mm to evaluate the models for robustness and determine trends in error scaling with increasingly noisy input data. This is represented mathematically by:

$$\mathbf{x} \rightarrow \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} + \begin{pmatrix} \text{rand}[-a, a] \\ \text{rand}[-a, a] \\ \text{rand}[-a, a] \end{pmatrix} = \tilde{\mathbf{x}} \tag{2}$$

where rand is an operator that selects a random value in the distribution and $\tilde{\mathbf{x}}$ is the noise injected into the testing data. We performed a sensitivity study to determine the optimum ANN hyperparameters for the final model, detailed in Appendix B and summarized in Table 2. The final model architecture, schematically drawn in Figure 5, is composed of 3 hidden layers, each composed of 128 neurons. The input is the 9-dimensional position data of the baseball location at consecutive frames in time $(\boldsymbol{x}_1, \boldsymbol{x}_2, \boldsymbol{x}_3)$ and the output is a 2-dimensional array of spin rate and velocity magnitudes, $\boldsymbol{y} = (|\omega|, \mathrm{v}_0)$. We utilized the adaptive moment estimation method (ADAM) for stochastic optimization (Kingma and Ba 2014) to minimize the mean-squared error, MSE, loss function given by:

Table 2: Summary of optimum hyperparameters determined through sensitivity study for use in final ANN training.

Batch size	Activation function	Hidden layers	Epochs
20	ReLu	3	100

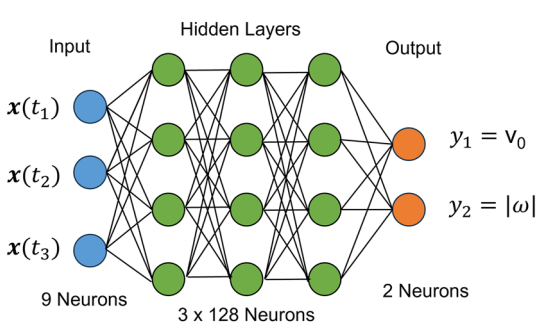


Figure 5: Final artificial neural network architecture composed of 9 input neurons, 3 hidden layers of 128 neurons (with ReLu activation), and 2 output neurons (with sigmoid activation). Input: $[\mathbf{x}(t_1), \mathbf{x}(t_2), \mathbf{x}(t_3)]$, Output: $\boldsymbol{y} = [\mathrm{v}_0, \omega]$.

$$MSE = \frac{1}{n}\sum_{i=1}^{n} (\mathbf{y_i} - \hat{\mathbf{y}}_i)^2 \tag{3}$$

where $\hat{\mathbf{y}}_i$ and $\mathbf{y}_i$ are the ANN predicted output and true values, respectively. We applied the sigmoid activation function, $\sigma(\zeta)$, on the output layer to return a normalized value of [0,1], represented mathematically as:

$$\sigma(\zeta) = \frac{1}{1 + \mathrm{e}^{-\zeta}} \tag{4}$$

where ζ is a scalar value. We trained the final ANN for 100 epochs, which was found to be a good compromise between training and testing loss to ensure the model was not over- or under-fit. Training of the ANN models on 24,000 pitch trajectories required approximately 5 min on a single Intel®Core™i7-8550U CPU.

2.3 Robustness and uncertainty analysis

We evaluated a relative and Euclidean error distance to quantify reconstructed pitch accuracy. All error measurements were taken from the midpoint of the pitch trajectory, consistent with previous approaches (Aguirre-López et al. 2018). The relative error, ϵ, is defined as the normalized difference in final position:

$$\epsilon = \frac{|\hat{\mathbf{x}} - \mathbf{x}_T|^2}{\mathbf{x}_T^2} \tag{5}$$

where $\hat{\mathbf{x}}$ is the position predicted by the ANN and $\mathbf{x}_T$ is the "true" position of the synthetic trajectory. Next, we defined the Euclidean distance in accordance with the work of Aguirre-López et al. (2018), allowing direct performance comparisons between the traditional iterative and ANN approaches. Further, it provides a more straightforward metric that can be directly compared to pixel error in camera tracking systems (Nickels and Hutchinson 2002). The Euclidean distance error, δ, is defined as the L2 norm of the distance between the true trajectory and ANN predicted trajectory:

$$\delta = \|\hat{\mathbf{x}} - \mathbf{x}_T\|_{L2} \tag{6}$$

We assessed the robustness of each ANN model (**Quiet** and **Noisy**) by calculating each error metric with increased injection of noise. This effectively simulates an increasingly noisy real-world camera tracking system. We injected error by sampling from a uniform distribution $[-a, a]$ in 2 mm increments up to a maximum error of 50 mm for each spatial coordinate.

We quantified the propagation of ANN model uncertainty in pitch prediction (input position uncertainty to output uncertainty) using the Monte Carlo method (MCM)

Table 3: Summary of cases considered for Monte Carlo method (MCM) uncertainty analysis (error is injected into the testing dataset).

Case	ANN model	Error injection
MCM-Q-2	**Quiet**	2 mm
MCM-Q-20	**Quiet**	20 mm
MCM-N-2	**Noisy**	2 mm
MCM-N-20	**Noisy**	20 mm

(Metropolis and Ulam 1949; Rubinstein and Kroese 2016). We chose a single representative pitch with a spin rate of 1,500 rpm and initial velocity of 90 mph. Then we injected noise to the trajectory using $a = 2$ mm and 20 mm assuming a uniform distribution, $[-a, a]$. This assessment simulates a comparison between a theoretical high-fidelity and low-fidelity camera tracking system, respectively. We define a cost function J as the standard error (SE) of the sampled distribution:

$$J = \mathrm{SE} = \frac{s}{\sqrt{n}} \tag{7}$$

where s is the sample standard deviation and n is the number of samples. We continued MCM sampling until convergence of the standard error of the model output distribution met a specified tolerance, resulting in approximately 20,000 samples for each distribution. The resulting probability distributions functions (PDF) were used to quantify the mean, standard deviation, and 95 % confidence interval of the model output. Table 3 shows a summary of the cases considered for uncertainty quantification. The ANN model output spin rate, initial velocity, and Euclidean distance error, are assessed in the MCM results.

3 Results

3.1 ANN model performance

The training and testing loss (MSE) is shown in Figure 6 for the **Quiet** and **Noisy** ANN models. The training and testing loss are closely correlated, as both sets are comprised of similar data from the same batch simulations. The **Quiet** ANN model is able to predict the pitch spin rate and velocity with nearly an order magnitude higher accuracy for both the training and testing datasets in comparison to the **Noisy** ANN model. The **Quiet** ANN model converges on a more precise mapping, due to clean high-fidelity training input. The **Noisy** ANN model has lower accuracy compared to the **Quiet** ANN model, as expected given lower-fidelity training data. Table 4 summarizes the final training and testing loss for both ANN models, which both demonstrate good convergence.

We reconstructed pitch trajectories to quantitatively assess ANN performance. Given the three spatiotemporal frames of position ($\mathbf{x}(t_1)$, $\mathbf{x}(t_2)$, $\mathbf{x}(t_3)$), each ANN model outputs the initial velocity and spin rate magnitudes (v_0, ω). These are fed as initial conditions to Equation (1) and used to reconstruct trajectories for all 6000 pitches in the testing dataset. The results are then compared to the true trajectories. Figure 7 shows a comparison between true and predicted trajectories for a single representative pitch with quantification of Euclidean error. The true, **Quiet** ANN model reconstructed, and **Noisy** ANN model reconstructed trajectories are shown in black, red, and gray, respectively. The y and z deflections are shown by the dotted and solid lines, respectively. Qualitatively, there is almost no

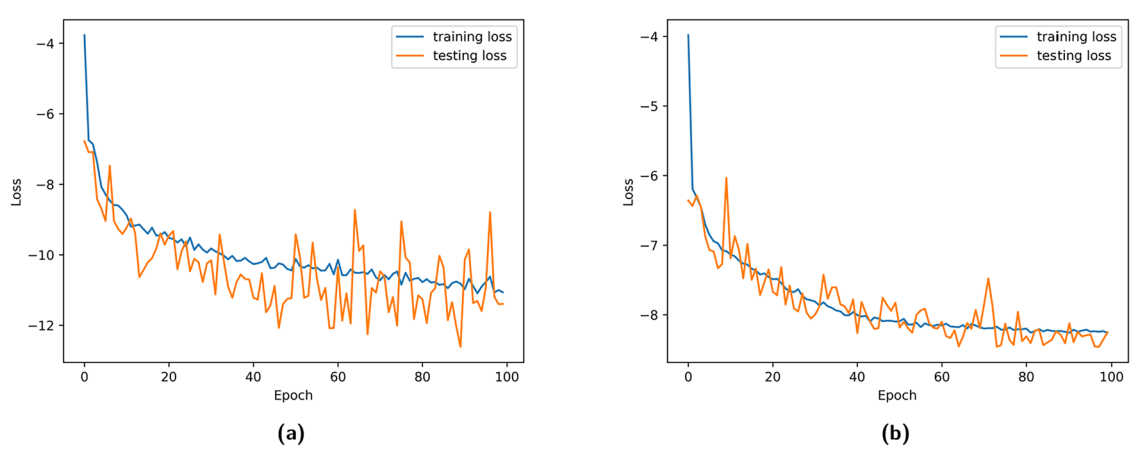


Figure 6: Training and testing MSE (defined in Eq. (3)) versus Epoch. a) **Quiet** and b) **Noisy** ANN models trained to predict the release speed (v_0) and spin rate $|\omega|$.

Table 4: Final training and testing MSE (defined in Eq. (3)) for the converged ANN model using 24,000 training and 6,000 test trajectories.

Model	Training loss	Testing loss
Quiet	1.51e-05	2.97e-05
Noisy	2.67e-04	2.69e-04

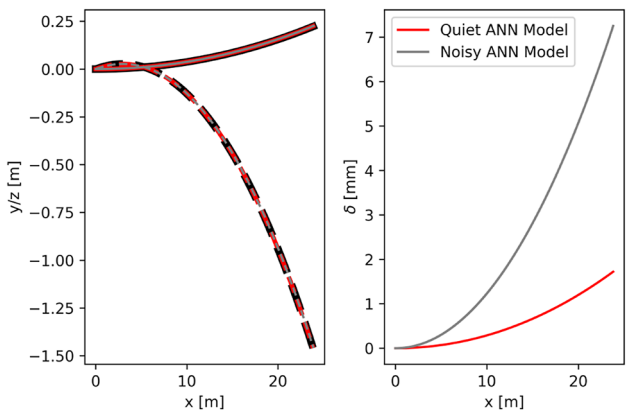


Figure 7: Comparison between true and predicted trajectories for representative pitch with Euclidean error comparison (no noise injected in testing data). **Quiet** ANN model outperforms **Noisy** model, with error accumulating along the pitch trajectory. Legend: (dotted: y, solid: z, black: True, red: **Quiet** ANN predicted, gray: **Noisy** ANN predicted).

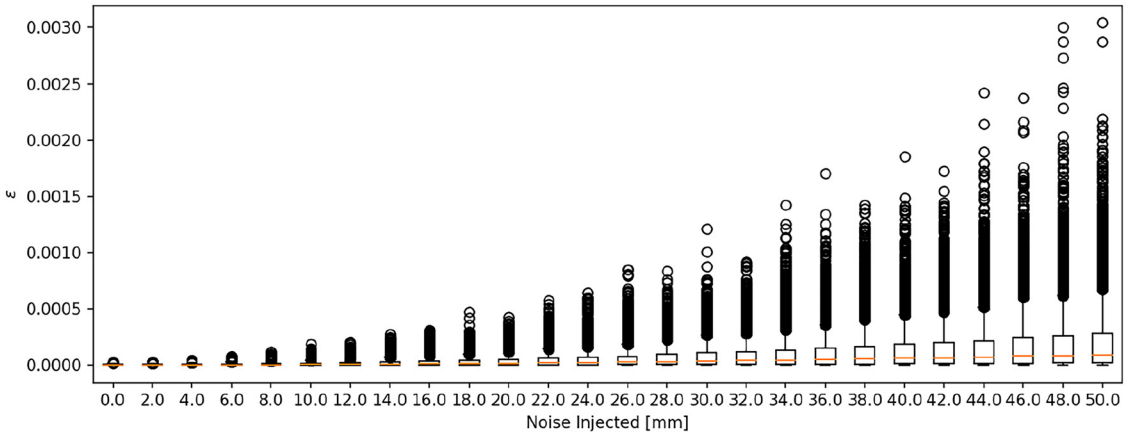
distinction between the simulated and the true pitch trajectories, showcasing the excellent performance of the models in pitch reconstruction. The performance of both ANN models suggests that the ANN model architecture is suitable for capturing the nonlinear relationship between position location and initial velocity and spin rate magnitudes utilizing the three-point approach. The evaluation of pitch trajectories using the ANN-based approach is highly computationally efficient compared with iterative physics-based methods (Aguirre-López et al. 2018). Once trained, the ANN provides the required initial conditions in a single forward pass, followed by one integration of the governing equations to determine the full trajectory. In contrast, traditional iterative solvers require multiple iterations of the governing equations to reach convergence, making them significantly more computationally expense. The main bulk of computational cost is handled *a priori* by the requirements for model training and generation of synthetic pitch data. After training, however, the ANN models are more computationally efficient than existing iterative approaches using the three-point reconstruction (Aguirre-López et al. 2018) which require multiple iterations, (typically >25), to converge on the pitch trajectory.

To assess the accuracy associated with a real-world noisy camera-tracking system, we inject increasing error into the testing dataset ($a = 2$ mm to $a = 50$ mm). Figures 8 and 9 show the relative and Euclidean error distributions with increased injected noise for both ANN models, respectively. The first major trend is that the injection of noise into the training dataset (**Noisy** ANN model) does appear to increase the robustness of subsequent model performance. For both the relative and Euclidean errors, the **Noisy** ANN model shows a fairly steady distribution of error below the training noise injection of 25 mm. Beyond 25 mm noise injection, the **Noisy** ANN model median and maximum error increase. This indicates that higher fidelity testing data, with noise injected below the training threshold ($a < 25$ mm), has little effect on the accuracy of model prediction. Model performance degradation only starts beyond this injection threshold, as lower fidelity input data ($a > 25$ mm) is outside the training limits. For the **Quiet** ANN model, there is a clear trend in increasing error with increasing noise in the testing dataset. The relative error follows a quadratic trend, while the Euclidean error has a linear trend (linearized due to the L2 norm operation). The maximum error for both metrics in the **Quiet** ANN model quickly surpasses the maximum error of the **Noisy** ANN model after injection of just 4 mm of noise. The model suffers detrimentally with increasing injection of testing noise, reaching predicted Euclidean errors roughly 4 times larger than the **Noisy** ANN model. Both models show a larger increase in maximum error with injected noise rather than median error, to be expected with random noise injection.

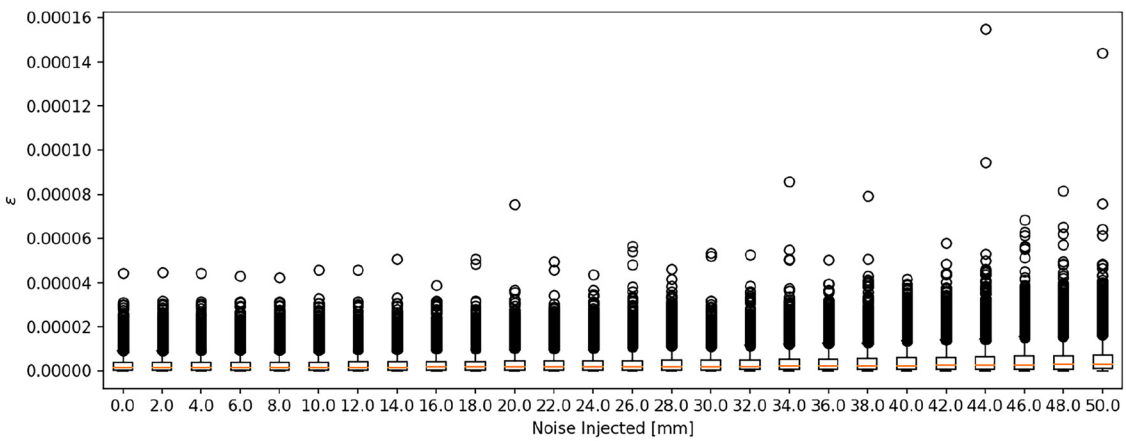
Table 5 summarizes ANN model performance without injected noise for both error metrics. There seems to be an incongruity in the results as the mean relative and Euclidean errors are lower for the **Noisy** ANN model in comparison to the **Quiet** ANN model. We reconcile this as a few data points with high error (outliers) disproportionately influence the MSE loss used to train the ANN models (the **Quiet** ANN model has a lower training loss than the **Noisy** ANN model). The **Quiet** ANN model Euclidean error distribution is the only distribution with no outliers in the testing evaluation, indicating that the model has optimized MSE loss by minimizing the influence of any spurious outlier data. However, the injection of noise in the training set forces the **Noisy** ANN model to optimize for a skewed distribution of error, allowing for outliers at the expense of improved overall resiliency to random noise.

3.2 Robustness analysis

For the application of low-cost pitch tracking, we expect low-fidelity input data. The computer vision and tracking



(a)



(b)

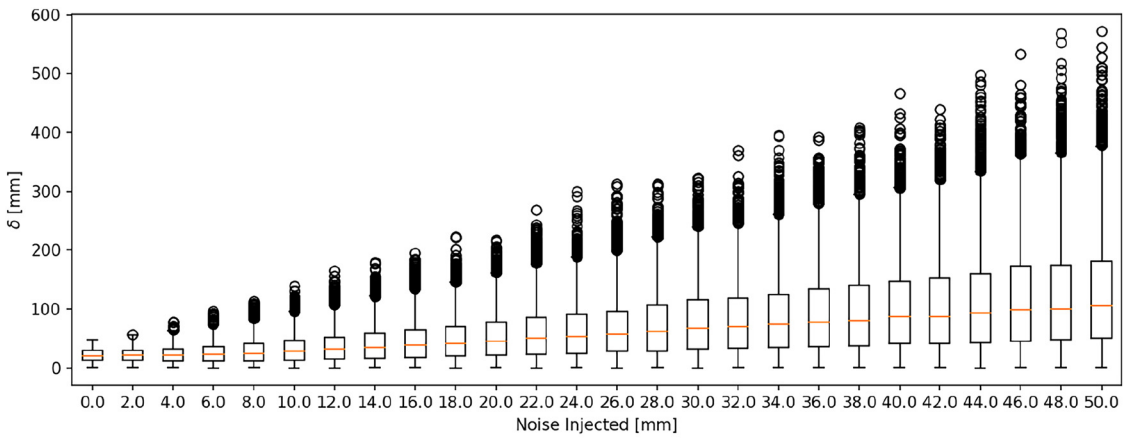
Figure 8: Relative error distributions with increasing noise in testing dataset. There is a quadratic increase in error for the a) **Quiet** ANN model, compared to a steady error for the b) **Noisy** ANN model with increasing noise injection. (Note: *y*-axis scales adjusted for better readability).

system introduces error in spatiotemporal location, arising from the quantity of cameras and quality (pixel density, frame rate, shutter-speed, calibration, etc.) of the system. Thus, quantifying the model robustness to various levels of input error is important for determining real-world applicability of the ANN models. Further, robustness quantification serves to determine the propagation of uncertainty to the output prediction (associated with the various levels of input uncertainty). We use a simple linear regression fit on the mean and maximum Euclidean error with increasing noise to compare performance of the ANN models. The linear model takes the form:

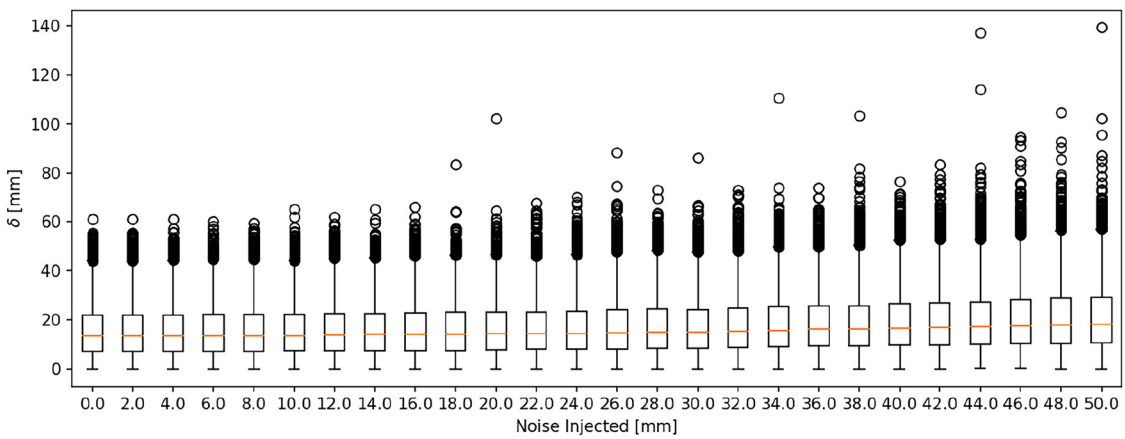
$$y = \hat{a}x + \hat{b} \tag{8}$$

where $\hat{a}$ and $\hat{b}$ are the linear regression fit the slope and y-intercept. For models with good linear fits, high R^2, the model provides a map for predicting uncertainty in the output (y) from input uncertainty (x), in the present linear model, the slope represents the marginal increase in prediction uncertainty with input uncertainty, and the y-intercept represents the systematic error present regardless of input uncertainty.

We also use the calculated Euclidean error to directly compare with the three-point reconstruction algorithm developed by Aguirre-López et al. (2018), which utilizes an iterative Newton–Raphson approach to minimize the error between simulated trajectories and given pitch locations. Aguirre-López et al. (2018) consider the same independent injection of uniform noise up to 2 mm in each coordinate direction. The iterative algorithm shows excellent agreement between simulated trajectory and target trajectory, with systematic error approaching zero for data with no noise injection. Considering error calculated at the midpoint



(a)



(b)

Figure 9: Euclidean error distributions with increasing noise in testing dataset. There is a linear increase in error for the a) **Quiet** ANN model, compared to a steady error for the b) **Noisy** ANN model with increasing noise injection. (Note: *y*-axis scales adjusted for better readability).

Table 5: Summary of relative error (Eq. (5)) and Euclidean error (Eq. (6)) metrics for ANN models tested without injected noise.

Model	Mean relative error	Max relative error	Mean Euclidean error	Max Euclidean error
Quiet	4.61e-06	2.69e-05	21.04	47.50
Noisy	2.86e-06	4.44e-05	15.62	61.04

of the trajectory, Aguirre-López et al. (2018) describe the trend in maximum Euclidean error prediction to be approximately linear, with a slope $\hat{a} \approx 3.5$ and intercept of $\hat{b} \approx 0$. We consider error injection of up to 50 mm, likely beyond the linear extrapolation of the iterative approach; however, the linear fit serves as a baseline for comparing the iterative approach to the present ANN model.

We show linear regression fits for the Euclidean error for both ANN models in Figure 10. We provide details of the model fits parameters and coefficients of determination in Table 6. We observe excellent agreement in the linear regression fit ($R^2 > 0.99$) for the **Quiet** ANN model maximum and average Euclidean error with increasing noise (see Figure 10.a). The **Noisy** ANN model (in Figure 10.b) shows good agreement in the average error ($R^2 = 0.939$), but the large variance in maximum error results in a weaker agreement between the regression model and error distribution ($R^2 = 0.581$). The poor agreement likely stems from the largely flat error prediction below 25 mm, where the injected testing noise is lower than the noise in the training set. Ultimately, the quality of pitch reconstruction (and subsequent collection of advanced analytics) is limited by the fidelity of the given camera tracking system. The injection of training noise in real-world deployment of an ANN

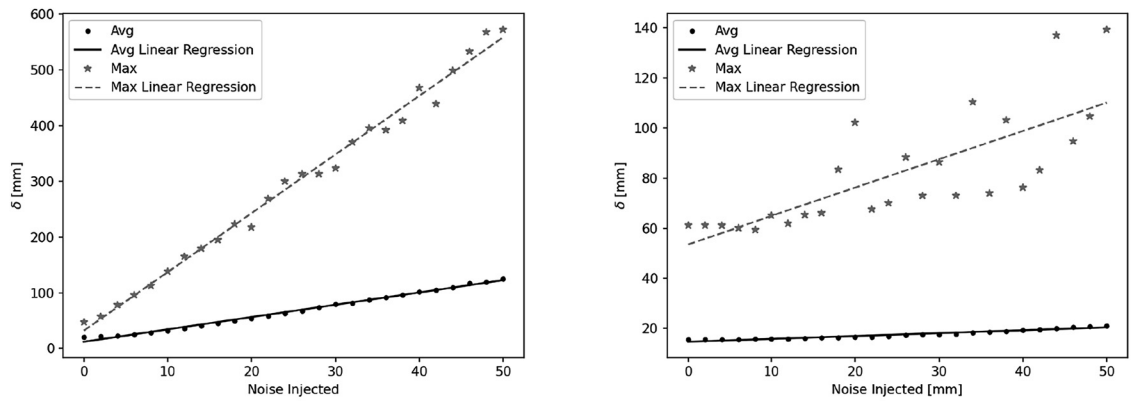


Figure 10: Linear regression fits for average and maximum Euclidean error by ANN model with increasing noise injection. The **Noisy** ANN model shows improved accuracy with increasing noise injection. (a) **Quiet** ANN model. (b) **Noisy** ANN model.

Table 6: Summary of linear regression fits for average and maximum Euclidean error by ANN model. The **Noisy** ANN model shows improved robustness to increased noise injection.

Model	Error metric	$\hat{a}$	$\hat{b}$	R^2
Quiet	Average	2.21	12.09	0.993
Noisy	Average	0.11	14.74	0.939
Quiet	Maximum	10.52	31.98	0.991
Noisy	Maximum	1.13	53.58	0.581

system should be carefully selected based on the expected maximum error of the camera tracking system.

We confirm the use of injected noise to increase the robustness of the ANN model by many trends seen in the linear regression model. Comparing the **Noisy** and **Quiet** ANN models, for both the average and maximum Euclidean error there is a substantial decrease in the predicted $\hat{a}$ value for the **Noisy** ANN model. This clearly indicates that the marginal increase in maximum uncertainty of the **Noisy** ANN model with increasing uncertainty in the input data is lower in comparison to the **Quiet** ANN model. The weaker dependence on predicted error with increasing input error in the system demonstrates the increased robustness of the **Noisy** ANN model. As noise is increased in the dataset, an ANN model trained with injected noise will quickly outperform an ANN model trained with clean high fidelity data. This greatly enhances the ANN model utility for the use case of pitch reconstruction on low-fidelity input data.

A drawback of injecting noise in the training dataset is that it increases the systematic error in the system. This can be observed by the increase in the y-intercept of the linear regression fits, $\hat{b}$. The rise in systematic error indicates that an ANN model with injected noise will not outperform an ANN model without injected noise at the limit of high-fidelity input data. However in comparison to the work of Aguirre-López et al. (2018), the **Quiet** ANN model performs poorly with no injected testing noise. The iterative model has a systematic error that approaches zero with perfect input data (Aguirre-López et al. 2018). This can be easily understood, as the ANN model is divorced from the governing physics of baseball motion. The iterative approach solves the baseball trajectory, using up to 25 trajectory simulations (Aguirre-López et al. 2018), directly from the governing equation during the pitch reconstruction, allowing for high precision results. Alternatively, the ANN model does not perform any simulations of pitch trajectory during pitch reconstruction, drastically reducing the computational cost. This separation from the ground-truth governing physics results in an increased uncertainty in the non-linear map between input pitch location and the initial pitch conditions.

Although the ANN models require less computational cost over the iterative method, their utility requires demonstration of improved performance and robustness. The maximum Euclidean error regression for Aguirre-López et al. (2018) has a slope $\hat{a} \approx 3.5$ and intercept of $\hat{b} = 0$. The linear regression for the **Noisy** ANN model shows a slope of 1.13 for the maximum Euclidean error, indicating a higher robustness to error than the traditional iterative methodology. The maximum Euclidean error is directly compared with the iterative approach in Figure 11. At 26 mm of injected noise (approximately 1 inch) the ANN model with injected noise is able to predict the baseball trajectory with higher accuracy than the traditional iterative approach. The ANN linear regression fit is actually able to predict the cross-over point where the ANN model outperforms the iterative approach of Aguirre-López et al. (2018) fairly well, predicting roughly 22 mm.

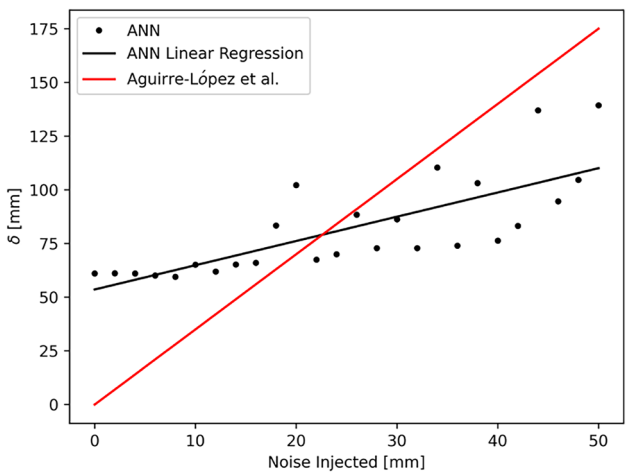


Figure 11: Comparison of maximum Euclidean error for **Noisy** ANN model and iterative approach (Aguirre-López et al. 2018) with increasing noise injection. The **Noisy** ANN model shows better robustness to noise beyond 25 mm.

3.3 MCM uncertainty analysis

The MCM quantifies the propagation of input error uncertainty in the ANN model. We assess prediction uncertainty for a single representative pitch with initial velocity, $v_0 =$ 90 mph and spin rate, $\omega = 1{,}500$ rpm. Figure 12 shows a comparison of the resulting PDFs quantifying prediction uncertainty between the **Quiet** and **Noisy** ANN models. Results show the Euclidean distance error, initial velocity, and spin rate predicted by each model with the injection of 2 mm and 20 mm of random uniform noise to the model input. As demonstrated previously, the **Quiet** ANN predicts lower Euclidean distance error and more accurate pitch velocity and spin rate in comparison to the **Noisy** ANN with the injection of low input noise (i.e. high-fidelity tracking system). The prediction PDF for the **Quiet** ANN is centered on the true spin rate and nearly centered on the true initial velocity, off by roughly 0.05 mph. With increased injection of noise, the predicted PDF widens appreciably, but remains centered on the true initial velocity and spin rate. In comparison, the **Noisy** ANN predicted PDF is not centered on the true initial velocity or spin rate, but the predicted PDF does not widen appreciably with increased injection of noise. The result is that the Euclidean error does not increase appreciably for the **Noisy** ANN as compared to the **Quiet** ANN, which sees much higher prediction inaccuracy. Interestingly, for both cases the injection of uniform noise results in a PDF that is largely normal, indicating the complexity of the representation held within the ANN models.

The results of the MCM study are quantified in Table 7 which shows the mean, standard deviation, and 95 % confidence interval for each PDF. We calculate the 95 % confidence interval by fitting and integrating the area of each resulting PDF using kernel density estimation. The **Quiet** ANN model consistently outperforms the **Noisy** ANN model in predicting the mean spin rate and velocity. In contrast, the **Noisy** ANN model is more tolerant to input uncertainty and the standard deviation rise is much lower between the high-fidelity and low-fidelity input. However, the **Noisy** ANN model is inaccurate with high-fidelity data, as the predicted output 95 % confidence interval does not contain the true initial spin rate or initial velocity. With increased input error the PDF is less affected, thus reducing the overall Euclidean error prediction.

These results demonstrate that injected noise in the input training dataset (**Noisy** ANN model) increases the robustness of the predicted value. This can be understood as the **Noisy** ANN model has learned a lower-fidelity representation of the baseball trajectory physics that is fault tolerant due to the presence of noise within the dataset. Alternatively, the **Quiet** ANN model does not contend with noisy training inputs and learns a higher-fidelity representation of the baseball trajectory physics, as evidenced by the lower training loss. Selection of ANN should be based on the available camera tracking system, and a proper estimation of camera tracking uncertainty should be established. This is beneficial both for 1) selection of noise injection in model training and 2) providing adequate uncertainty metrics for ultimate use in data analysis.

The use of ANNs represents an attractive alternative approach to pitch reconstruction suitable for fast reconstruction with insensitivity to injected noise. Practical implementation of this pitch reconstruction algorithm requires integration with pitch tracking systems. Standard ball tracking systems, such as Hawk-Eye (Collins and Evans 2008), utilize a series of highly accurate cameras calibrated together to provide position data with each frame. By decreasing the requirement on data fidelity, this approach could utilize standard cameras to provide the low-fidelity spatiotemporal data. Further, by requiring only three frames, the memory requirement would be insignificant compared to existing systems. The spatiotemporal frames from the camera tracking system would be fed into the ANN reconstruction methodology to generate advanced pitch trajectory analytics. The low total cost of the system makes it suitable for implementation at the amateur level and international stages, empowering smart-training and improving player scouting.

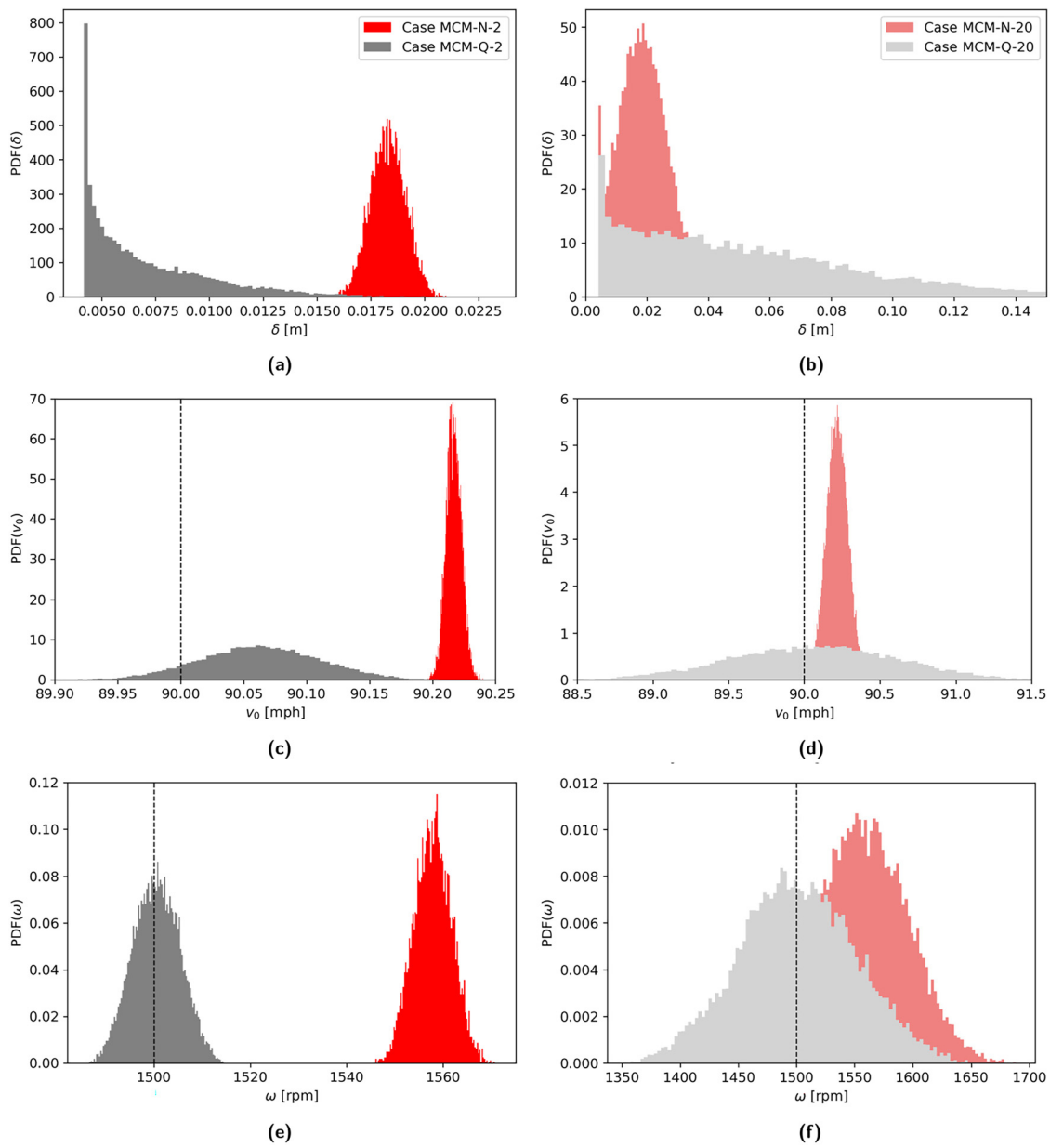


Figure 12: Comparison of probability distribution functions (PDFs) of prediction uncertainty between **Quiet** and **Noisy** ANN models using MCM (Vertical lines are actual values: $\omega = 1{,}500$ rpm $v_0 = 90$ mph). The **Noisy** ANN model performs poorly with 2 mm noise injection, but outperforms the **Quiet** ANN model with 20 mm noise (especially evident by the Euclidean error measurement). (a) Euclidean error, 2 mm noise injection. (b) Euclidean error, 20 mm noise injection. (c) Initial velocity, 2 mm noise injection. (d) Initial velocity, 20 mm noise injection. (e) Spin rate, 2 mm noise Injection. (f) Spin rate, 20 mm noise injection.

Table 7: Summary of prediction uncertainty distributions using MCM. Mean, standard deviation, and 95 % confidence intervals calculated for initial velocity, spin rate, and Euclidean error.

Case	ANN model	Error injection	Parameter	Mean	Standard deviation	95 % CI
MCM-Q-2	**Quiet**	2 mm	v_0 [mph]	90.1	0.06	[89.95, 90.17]
			ω [rpm]	1,501	4.84	[1,491, 1,510]
			δ [mm]	7.1	2.96	[3.4, 14.9]
MCM-Q-20	**Quiet**	20 mm	v_0 [mph]	90.1	0.54	[88.96, 91.12]
			ω [rpm]	1,501	49.74	[1,402, 1,600]
			δ [mm]	51.6	37.10	[6.4, 140.4]
MCM-N-2	**Noisy**	2 mm	v_0 [mph]	90.22	0.01	[90.20, 90.23]
			ω [rpm]	1,558	3.86	[1,550, 1,566]
			δ [mm]	18.3	0.81	[16.7, 19.9]
MCM-N-20	**Noisy**	20 mm	v_0 [mph]	90.2	0.07	[90.07, 90.36]
			ω [rpm]	1,558	37.45	[1,486, 1,634]
			δ [mm]	18.6	7.64	[5.4, 34.2]

4 Conclusions

Widespread detailed pitch tracking is currently infeasible due to the limiting high cost of systems employed by MLB and high level college baseball. Advances in pitch tracking (which provides quantitative data on pitch speed, movement, and spin) are necessary to enable better decision making for personnel managers. We investigate the use of artificial neural networks and a three-point pitch reconstruction scheme as a possible method for developing cost-effective baseball pitch tracking systems where high-end technology is inaccessible. We demonstrate that ANNs trained using synthetic pitch trajectories can successfully predict the spin rate and velocity with high accuracy.

Two ANN models were trained, one with high-fidelity training data (**Quiet**) and one with uniform noise of 25 mm injected into the training dataset (**Noisy**). We then injected uniform noise, in increments of 2 mm up to a maximum of 50 mm, into the testing dataset to evaluate the performance of the ANN models (simulating low-fidelity camera integration).

Both ANN models were able to reconstruct pitch trajectory with good accuracy. We performed a robustness analysis, demonstrating improved robustness qualities of the **Noisy** ANN model over the **Quiet** ANN model. We calculated probability distributions functions (PDFs) of predicted model output using the MCM to quantify model uncertainty with increased noise. The **Quiet** ANN model outperformed the **Noisy** ANN model with high-fidelity testing data, but quickly under-performs with increasing uncertainty in the testing dataset. The **Noisy** ANN model learned a lower-fidelity representation of the governing physics, resulting in a large increase in standard deviation for the **Quiet** ANN model with increasing error. In comparison to traditional iterative methods, the **Noisy** ANN model had less dependence on error and improved robustness properties. We demonstrated that the ANN-based model is suitable for low-fidelity noisy input, providing an alternative means for cost-effective pitch reconstruction.

Future work is required before the model is suitable for real-world application. To fully reconstruct pitch trajectory, the model must be able to predict the axis of rotation and the launch angle of the pitch, requiring an increase in the number of model parameters. Further, the model assumes a known release point and time relative to release, which will add further uncertainty in spatiotemporal pitch location. Robustness to semi-random or directional noise requires additional testing to ensure that the system can tolerate systematic errors in low-cost camera tracking. The ANN model also requires integration with real-world tracking systems to compare directly with traditional pitch reconstruction approaches. The algorithm can be extended to reconstruct hitting trajectory; and with modification to the governing equations, to other sports trajectory reconstructions.

Research ethics: Not applicable.
Informed consent: Not applicable as there were no human subjects and IRB approval is not applicable.
Author contributions: R. DeBoskey – Conceptualization, Methodology, Software, Visualization, Writing – original draft, Writing – reviewing and editing. V. Hasti – Conceptualization, Methodology, Software, Resources, Supervision, Writing – reviewing and editing. V. Narayanaswamy – Supervision, Resources, Writing – reviewing and editing.
Use of Large Language Models, AI and Machine Learning Tools: None declared.
Conflict of interest: The authors declare that they have no known competing financial interests or personal

relationships that could have appeared to influence the work reported in this paper.

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Data availability: Data will be made available upon reasonable request.

Appendix A: Baseball trajectory model

Equation (1) is the governing equation for baseball trajectory, derived from Newton's second law of motion. The equation is a simplified version derived from Aguirre-López et al. (2018), which assumed the centrifugal and Coriolis body forces are both negligible. The centrifugal and Coriolis forces are approximately two orders of magnitude smaller than the magnus force and are generally negligible to trajectory calculation (Robinson and Robinson 2017). The drag force, $\mathbf{f}_D$, was given by Giordano (1996):

$$\mathbf{f}_D = -\left(0.0039 + \frac{0.0058}{1+\exp((\mathrm{v}-\mathrm{v}_d)/\Delta)}\right)\mathrm{v}\cdot\mathbf{v} \tag{9}$$

where $\mathbf{v}$ is the velocity vector, v is the velocity magnitude, and $\mathrm{v}_d = 35$ m/s and $\Delta = 5$ m/s are calibrated model constants. The magnus force, $\mathbf{f}_M$, is present and plays a prominent effect in the physics of ball trajectory in many sports (Kray et al. 2014; Sayers and Hill 1999). The force is oriented orthogonal to the spin axis and velocity, resulting in the well-known curving of baseball pitches. By Bernoulli's principle, the net force was given by the cross product ($\times$) of velocity about the spin axis:

$$\mathbf{f}_M = B(\boldsymbol{\omega}\times\mathbf{v}) \tag{10}$$

where $B = 0.00041$ is a non-dimensional scaling factor given by Giordano (1996) for baseballs. We assumed a constant acceleration due to gravity, $g = -9.81$ m/s^2, in the z-direction. We solved Equation (1) using 4th-order Runge–Kutta time integration (Lomax et al. 2001) with a fixed time step, $\Delta t = 2.083$ ms.

Several simplifying assumptions were made in the current work. First, we neglected the orientation of the baseball seams. Extremely complicated physics, including the seam-shifted wake phenomenon (Smith and Smith 2021), arise from the transition and break-up of the boundary that develops over the ball due to the presence and orientation of the seams on an otherwise smooth sphere (Higuchi and Kiura 2012). This results in small fluctuations to the baseball trajectory, which may not be negligible compared to magnus forces (Borg and Morrissey 2014; Higuchi and Kiura 2012; Smith and Smith 2021).

Secondly, the batch simulations assume a fixed spin axis, $\phi = 45°$, fixed launch angle $\alpha = 1°$, and constant release point, $\boldsymbol{x}_o = (0, 0, 0)$. In addition to the speed and spin rate, the spin axis is very important for pitch classification (the constant angles of $\phi = 45°$ and $\alpha = 1°$ used in the present study correspond roughly with a curveball from a right-hand pitcher). The magnus force is directed orthogonal to the spin axis and velocity (via Eq. (10)) which causes the breaking direction of the pitch. With differing amounts of break and velocity, the pitcher must adjust the launch angle accordingly between pitches. The fixed angles and constant release point were used in the present work to reduce dimensionality to demonstrate the use of an ANN as a proof-of-concept for pitch reconstruction. Higher dimensional prediction output (necessary to predict spin axis, launch angle, and initial condition) requires a much larger ANN model and substantially more training data (curse of dimensionality), which is outside the scope of the present work. The spatiotemporal release point of the baseball must be known to determine the relative deflection of the pitch. We assumed a known release point in this paper, which in practice must be determined by the system to fully reconstruct trajectory. This introduces an additional data point (and associated uncertainty) to collect and process within a real-world pitch tracking system.

Lastly, we sampled the training data for the ANN at a fixed constant interval ($t_1 = 0.467$ s, $t_2 = 0.483$, and $t_3 = 0.5$s). The selection of time interval in the present work was arbitrary, where t_3 is the (approximate) time required for an 80 mph pitch to reach home plate without consideration of the pitcher's extension from the rubber. Sampling at the same three time points (relative to release) cannot be realized in practice, particularly for a low-cost vision tracking system. The relative time from release would need to be varied for a real-world system, providing another layer of complexity.

Appendix B: ANN sensitivity study

We performed a sensitivity study to determine the training batch size, activation function for the hidden layers, and number of hidden layers to be used by the final model. Table 8 summarizes the range of hyperparameters tested in

Table 8: Summary of ANN hyperparameter sensitivity study.

Hyperparameter	A	B	C	D
Batch size	10	20	30	40
Activation function	sigmoid	tanh	relu	leakyrelu
Hidden layers	1	2	3	4

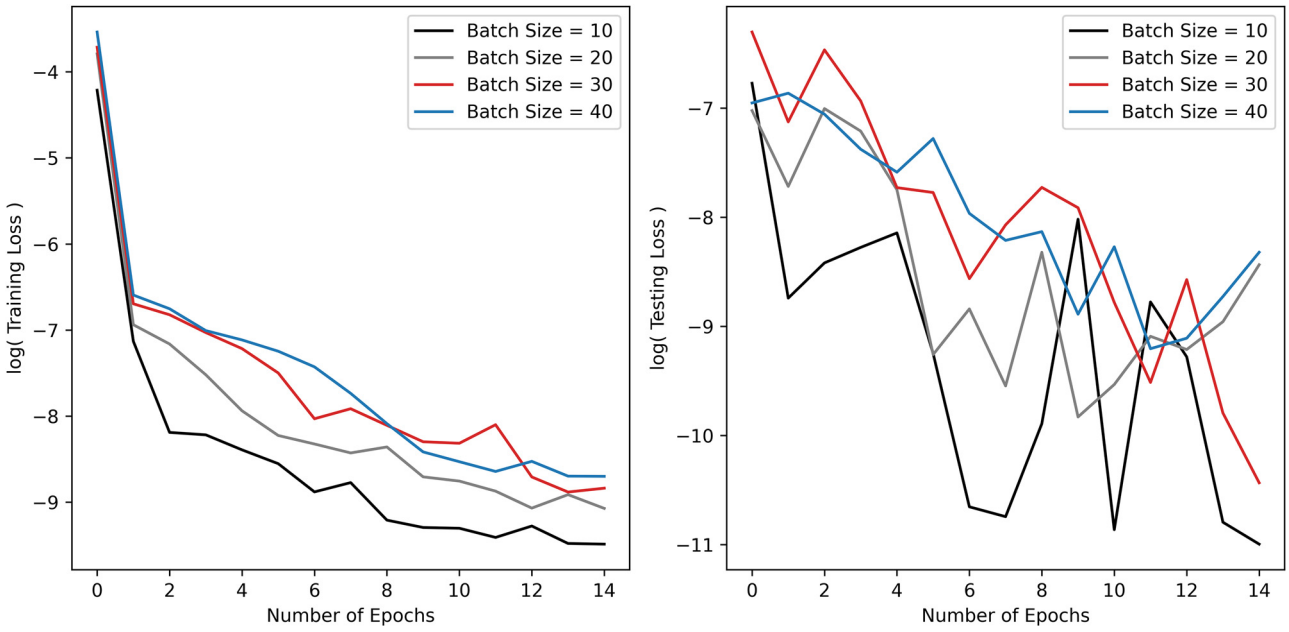

Figure 13: Sensitivity study on batch size showing its effect on the training and testing MSE loss (Eq. (3)) versus epoch for the **Quiet** ANN model.

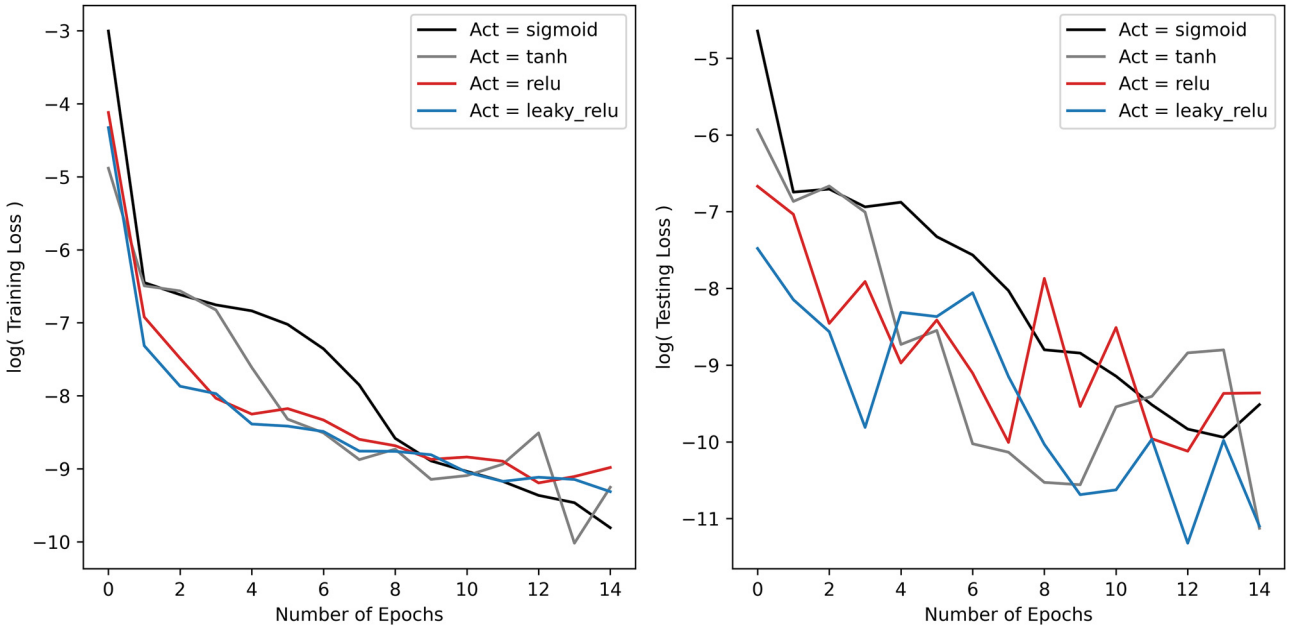

Figure 14: Sensitivity study on activation function showing its effect on the training and testing MSE loss (Eq. (3)) versus epoch for the **Quiet** ANN model.

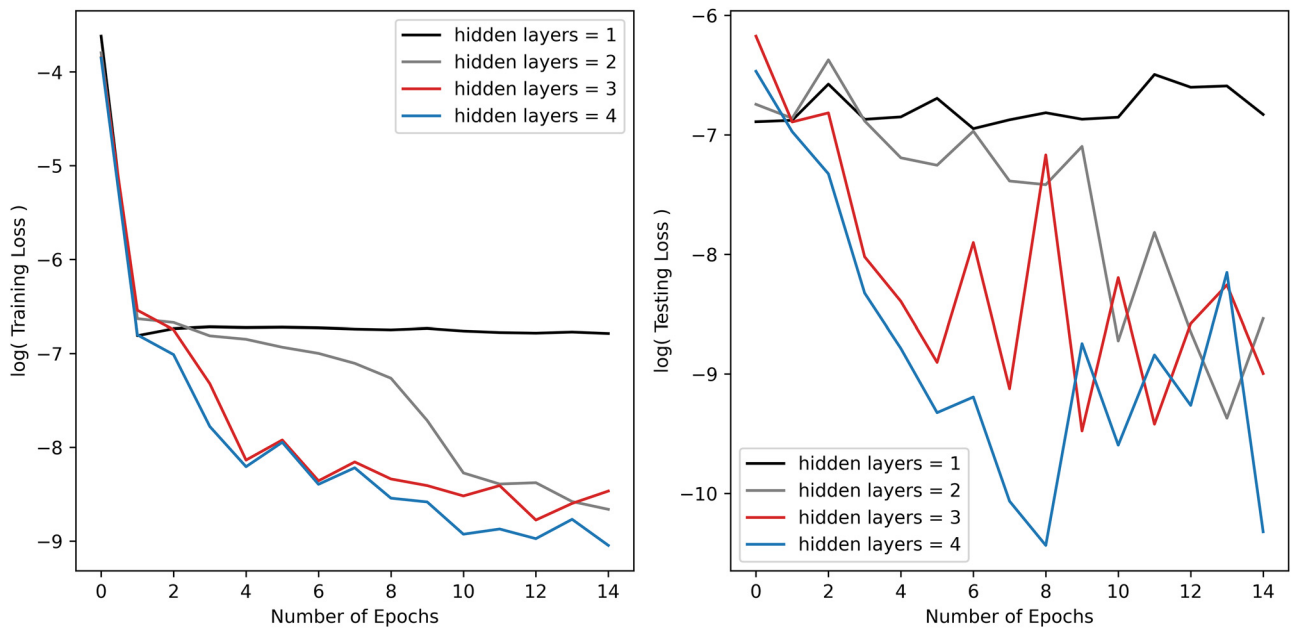

Figure 15: Sensitivity study on number of hidden layers showing its effect on the training and testing MSE loss (Eq. (3)) versus epoch for the **Quiet** ANN model.

this sensitivity study. We used minimization of the mean-squared training and testing loss as the parameter to evaluate the ANN performance. The sensitivity study was performed only on the **Quiet** ANN model, as a larger number of training epochs was required to train this model. The decrease in time required to train the **Noisy** ANN model was indicative of a less complex mapping due to lower-fidelity input data.

Figures 13–15 show the results of the sensitivity analysis for batch size, activation function, and number of hidden layers, respectively. For all plots, the logarithm of the testing and training loss was taken to better visualize the relative differences between model parameters. The ANN training time was largely insensitive to batch size. Although the loss decreases significantly with fewer epochs, a smaller batch size required more computation as the network parameters were updated more frequently. A batch size of 20 is taken in the final model architecture, as a compromise between number of epochs and speed of training. The ANN was insensitive to activation function and we used the ReLu activation function, $R(\zeta)$, in the hidden layers, represented mathematically as:

$$R(\zeta) = \max(0, \zeta) \tag{11}$$

The ANN requires 2 hidden layers or greater to accurately model the nonlinear relationship of velocity and spin rate. Once the model is larger than 2 hidden layers it was relatively insensitive to an increasing number of layers. To ensure that the model was able to capture the nonlinear trends, the final model architecture contains 3 hidden layers.

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Graphical synopsis

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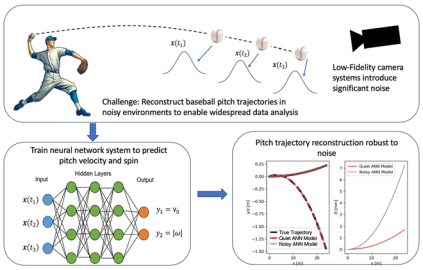
Robust baseball pitch reconstruction using artificial neural networks with noisy data

https://doi.org/10.1515/znb-2025-0057
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